

Raj Kumar

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RESEARCH INTERESTS

Physics-informed machine learning and probabilistic forecasting for solar energy: calibrated prediction intervals, multimodal forecasting with LLM-encoded operational context, cross-site generalisation (leave-one-site-out evaluation), and data augmentation for imbalanced solar-panel imagery.

EDUCATION

Jeju National University, Jeju, South Korea Mar 2025 – Feb 2027 (expected)
M.S. in Electronics Engineering, Machine Learning Lab

- **Advisor:** Prof. Yung-Cheol Byun · **Co-advisor:** Prof. Yong-Woon Kim
- **Proposed thesis:** Physics-Informed LLM-Augmented Multimodal Framework for Regime-Aware Probabilistic Solar Power Forecasting
- **Funding:** BK-21 fully funded scholarship

Sukkur IBA University, Sukkur, Pakistan Aug 2016 – Aug 2020
B.E. in Electrical Engineering (Electronics)

- **Final-year project:** AI-Based Plant Health Monitoring System

JOURNAL PUBLICATIONS

[J1] **Raj Kumar**, Yong-Woon Kim, Yung-Cheol Byun. "Mitigating Data Imbalance Using Image-Based Stable Diffusion and Feature-Level SMOTE for Solar Panel Classifications with CNNs." *Energy Reports*, 2026. doi:10.1016/j.egyr.2026.109055

[J2] **Raj Kumar**, Yong-Woon Kim, Yung-Cheol Byun. "Multi-Tier Data Augmentation and Balancing Framework Integrating Diffusion, Tomek Link, and SMOTE (DiToS) for Robust Image-Based Fault Detection in Solar Panels." *Sustainable Energy Technologies and Assessments*, 2026. doi:10.1016/j.seta.2026.104916

[J3] **Raj Kumar**, Yong-Woon Kim, Yung-Cheol Byun. "Hybrid Framework Combining Diffusion-Based Image Augmentation and Feature Level SMOTE for Addressing Extreme Class Imbalance." *IEEE Access*, 2025. doi:10.1109/ACCESS.2025.3600622

[J4] **Raj Kumar**, Gulsher Baloch, Pankaj, Abdul Baseer Buriro, Junaid Bhatti. "Fungal Blast Disease Detection in Rice Seed Using Machine Learning." *International Journal of Advanced Computer Science and Applications (IJACSA)*, 2021. doi:10.14569/IJACSA.2021.0120232

CONFERENCE PUBLICATIONS

[C1] **Raj Kumar**, Yong-Woon Kim, Yung-Cheol Byun. "Robust Cross-Site Solar Power Forecasting via Semantic Operational Feature Integration Under LOSO." *Korea Information Technology Society (KIIT) Summer Conference, Jeju*, 2026.

[C2] **Raj Kumar**, Yong-Woon Kim, Chan Yeob Yeun, Yung-Cheol Byun. "Multimodal Cross-Site Solar Energy Forecasting Using Large Language Model." *ICOBAR-SMART Joint Conference, BINUS University, Jakarta*, 2026. DBpia

[C3] **Raj Kumar**, Yong-Woon Kim, Nazrul Amin, Dong-Cheol Lee, Yung-Cheol Byun. "LLM-Augmented Multimodal Approach to Reliable Solar Energy Forecasting." *International Conference on Convergence Content (ICCC 2025), NTUT, Taipei*, 2025. DBpia

[C4] **Raj Kumar**, Yong-Woon Kim, Yung-Cheol Byun. "Image-Level Generative Oversampling and Boundary-Based Cleaning for Solar Panel Fault Detection." *International Conference on Advanced Information Technology (ADINTECH 2025), Phnom Penh*, 2025. DBpia

[C5] **Raj Kumar**, Yong-Woon Kim, Yung-Cheol Byun. "Hybrid Approach to Mitigate Extreme Class Imbalance Using Stable Diffusion and SMOTE." *Korea Information Technology Society (KIIT) Summer Conference, Jeju*, 2025. DBpia

Under review: as of September 2026, six further manuscripts are submitted to *Applied Energy*, *Renewable Energy*, *Knowledge-Based Systems*, *Applied Soft Computing* and *IEEE Access*.

HONORS & AWARDS

1st Position, Best Presentation — BK-21 10th Colloquium on New Energy Industry, JNU	Feb 2026
Best Paper Award — ICOBAR-SMART Joint Conference, Jakarta [C2]	Jan 2026
Best Presentation Paper Award — ICCC 2025, National Taipei University of Technology [C3]	Dec 2025
Best Paper Award — ADINTECH 2025, Phnom Penh [C4]	Jul 2025
BK-21 Fully Funded Scholarship — Jeju National University (M.S.)	Since 2025
LU Continental Scholarship — Sukkur IBA University (B.E.)	2016 – 2020
THP-2016 Scholarship — Sukkur IBA University (foundation semester)	2016

RESEARCH & PROFESSIONAL EXPERIENCE

- Research Scholar — Machine Learning Lab, Jeju National University, South Korea

Mar 2025 – Present

 - Research on physics-informed, multimodal and probabilistic solar power forecasting, fault detection and synthetic data generation for imbalanced datasets.
 - Built and deployed AI tooling used by the lab: an automated first reviewer for weekly meeting material (Slack) and a bilingual AI lab member (Telegram).
- Machine Learning Engineer — MasterMinds Enterprises, Karachi, Pakistan

Jun 2023 – Feb 2025

 - Led design and testing of ML systems for predictive maintenance and optimisation, integrated into larger engineering workflows.
- Research Engineer — Augmented Startups, Johannesburg, South Africa

May 2022 – Apr 2023

 - Computer-vision research and models for object detection, localisation and classification; curriculum support for self-driving-car topics.
- Research Assistant — Sukkur IBA University, Pakistan

Aug 2019 – Apr 2021

 - Machine learning and image processing for plant and human disease detection, using augmentation and transfer learning.

SELECTED PROJECTS

- PI-Solar LAB** — interactive research dashboard comparing physics-only, ML-only and physics-informed forecasters on 42 PV sites, with probabilistic metrics and significance tests (Streamlit, pvlib, PyTorch, XGBoost/LightGBM/CatBoost; 136 tests).
- LabReviewer** — Slack agent that drafts supervisor-style reviews of weekly lab material for professor approval; runs on a local vision-language model (LangGraph, Ollama/Qwen3-VL, vector search; 844 tests).
- Lora** — bilingual (Korean/English) AI lab member on Telegram that produces documents, slides and cited notes on request, with voice input and output (Claude Code, SQLite FTS5, faster-whisper; 96 tests).
- Scientific Animation Studio** — local dashboard that turns research into narrated explainer videos; deterministic renderers for equations and data, an open-weight video model for motion (Claude, Wan 2.2, Manim, Blender).
- Cross-Model Review Team** — public guide and workflow running Claude Code and Codex as reviewing teammates in Block's Buzz ([GitHub](#)).

INVITED TALKS & OTHER PRESENTATIONS

- Poster: Physics-Informed Probabilistic Solar Power Forecasting — Green Hydrogen Global Forum, Jeju

Sep 2026
- Talk: LLM-Augmented Multimodal Solar Energy Forecasting — BK-21 10th Colloquium on New Energy Industry, JNU

Feb 2026
- Invited lecture: Large language models for solar power forecasting — Global Hyperscale AI Camp, JNU

Jan 2026
- Talk: Hybrid Framework (DISMHA) for extreme class imbalance — Jeju-Philippines R&D Symposium, JNU

Jun 2025
- Talk: On-device AI for drone-based solar panel inspection — Science Immersion Program 2025, Jeju

Jun 2025

SERVICE & OUTREACH

- Peer reviewer — PLOS ONE

Sep 2025
- Article — pv magazine: "How to Address Imbalance Datasets in Solar Panel Dust Detection" ([link](#))

Mar 2026

TECHNICAL SKILLS

- Machine learning:** Python, PyTorch, XGBoost, LightGBM, CatBoost, pandas, quantile regression, conformal calibration, Stable Diffusion-based augmentation
- Energy & physics:** pvlib (solar position, clear-sky irradiance, PV performance models)
- LLMs & agents:** Claude Code, Codex, LangGraph, Ollama, Pydantic structured output, embeddings and vector search, tool use
- Engineering:** Streamlit, FastAPI, Git, GitHub Actions, pytest, Playwright, Linux GPU workstations